

Hachem Nasri

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PROFILE

Computer Science graduate building and evaluating blockchain protocols with Solidity, Noir, and Rust. Experience integrating browser-based zero-knowledge proofs, relayed transactions, and product authentication workflows, supported by executable tests and technical reports. Research interests include eligibility with less disclosure, Sybil resistance, and the trust boundaries of cryptographic authentication.

EDUCATION

Bachelor of Science in Computer Science

Graduated 2026

Institut Supérieur d'Informatique et des Technologies de Communication de Sousse, Tunisia

PROTOCOL PROJECTS

Private Airdrop

Repository

Private eligibility and relayed ETH rewards · Noir, Rust/WASM, Solidity, React, Node.js

- Built a browser-based proof workflow that keeps the registration wallet out of public proof inputs; a server relays the claim to a chosen recipient.
- Connected a Noir circuit to Rust/WASM helpers for hashing, input normalization, and 256-leaf Merkle trees, then integrated Solidity admin/user facets and a generated verifier.
- Developed **60 Noir, 6 Rust, and 23 Solidity tests** covering membership, signatures, replay rejection, permissions, and payouts, including real-verifier checks. Documented issuer linkability and the unresolved limits of repeated participation and Sybil resistance.

Provenance

Repository

Challenge–response authentication · Solidity, Hardhat, React/TypeScript, Express, MongoDB

- Built brand, retailer, and consumer workflows for custody transfers, returns, sales, and signature verification across six contract facets.
- Integrated a software key service to simulate the proposed PUF authenticator; made explicit that a valid signature does not establish the authenticator's physical connection to a genuine product.
- Evaluated contracts with **81 passing tests and 85.68% recorded line coverage**, including cases that expose replay and authorization limitations.

Decentralized Certificate System

Repository

Non-transferable credentials · Solidity, OpenZeppelin, Hardhat, React

- Built an issuer-controlled credential workflow with public certificate lookup and transfer rejection for wallet-bound certificates.
- Implemented issuer enrollment, status management, and role-based authorization.
- Added contract tests for permitted issuance and rejected operations, connecting access policies to the behavior enforced by the contract.

TECHNICAL SKILLS

Languages	Solidity, Rust, Noir, TypeScript, JavaScript
Cryptography & Protocols	Zero-knowledge circuits, Merkle membership proofs, digital signatures, nullifiers, role-based access control, contract state machines, Diamond-style routing
Development Tools	Foundry, Hardhat, ethers.js, OpenZeppelin, WebAssembly, React, Node.js, Express, MongoDB, Git
Evaluation	Circuit and contract testing, negative test cases, coverage interpretation, threat modeling, technical reports, Typst

SELECTED TECHNICAL WRITING

Private Airdrop: Wallet-Bound Membership Proofs for Relayed Rewards **2026 Report**

Project technical report

Explains the registration-to-claim protocol, circuit and contract responsibilities, test evidence, and privacy boundaries, including issuer linkability and the limits of duplicate-claim prevention.

Provenance: Product Traceability with a Simulated Hardware Authenticator **2026 Report**

Project technical report

Examines custody workflows and challenge–response verification, interprets contract test results, and distinguishes software guarantees from assumptions about physical products.